

ABDALLA ELSIDDIG

AI & Robotics Engineer | Computer Vision Specialist | Autonomous Systems

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PROFESSIONAL SUMMARY

Top 5 national finalist in the Huawei ICT Competition (Bahrain, 2025) and 1st place university robotics champion, I build and deploy autonomous AI systems at the intersection of computer vision, robotics, and embedded engineering. Final-year Informatics Engineering student with 1.5+ years of industry experience delivering production-grade systems — including a SLAM-based AMR for a national healthcare supplier and an AI rehabilitation platform for children with special needs. Now working as an AI Automation Engineer, designing intelligent automation workflows and AI-driven systems. Founder and President of UTB IoT Club (150+ active members), holder of 10+ certifications across AI, IoT, and cloud platforms.

KEY ACHIEVEMENTS

- **Top 5 National** — Huawei ICT Competition 2025 (Innovation Track: AI & Healthcare) representing University of Technology Bahrain
- **1st Place** — University Sumo Robot Competition: custom-engineered fully autonomous robot with advanced sensor integration
- **President & Founder** — UTB IoT Club scaled to 150+ active members over 3 years with structured engineering curriculum and 12+ workshops
- **Production Deployment** — SLAM-based AMR at Shifa National Medical Supply, reducing manual warehouse operations by 40%
- **Government Partnership** — FireX autonomous robot demonstrated at national fire festival with Ministry of Interior & Civil Defense

EDUCATION

Bachelor of Informatics Engineering

2022 – 2026 (Expected)

[University of Technology Bahrain](#) | GPA: 3.38 / 4.0

Relevant Coursework: Machine Vision, Robotics, Software Engineering, Networking, Power Electronics, Electronics Engineering, Ethical Hacking

PROFESSIONAL EXPERIENCE

AI Automation Engineer — Glam Moda

Apr 2025 – Present

- Design and deploy AI-driven automation workflows that streamline business operations, integrating LLMs, APIs, and no-code/pro-code tools to reduce manual effort across the organization.
- Build intelligent agents and automation pipelines connecting internal systems, data sources, and third-party platforms for end-to-end process automation.
- Develop custom AI solutions — from prompt engineering and model integration to production deployment — aligning technical implementation with business goals.
- Collaborate across teams to identify automation opportunities, prototype solutions rapidly, and deliver measurable efficiency gains.

President & Founder — IoT Club — [University of Technology Bahrain](#)

2023 – Present

- Founded and scaled UTB's IoT Club to 150+ active members, establishing the most active student engineering community at the university.
- Designed structured curriculum covering embedded systems, Arduino programming, ROS, and real-time sensor integration across 12+ hands-on workshops.
- Organized quarterly innovation challenges, competitive robotics events, and industry speaker sessions connecting students to professional opportunities.
- Mentored 40+ junior engineers in IoT design, embedded systems development, and competitive robotics from beginner to competition-ready level.

Robotics Workshop Instructor & Competition Lead — [University of Technology Bahrain](#)

2023 – Present

- Designed and delivered robotics engineering curriculum bridging embedded system theory with competitive and practical implementation.
- Won 1st place in University Sumo Robot Competition with custom-engineered autonomous robot featuring advanced sensor fusion and control algorithms.
- Mentored 20+ students in robotics engineering, embedded development, and competitive problem-solving methodologies.

IT Technical Support & Web Developer — Glam Moda

2024 – 2025

- Specialized in Shopify platform engineering: custom theme development, third-party app integration, and performance optimization achieving 40% improvement.
- Collaborated with development team across full software lifecycle: architecture design, implementation, testing, and deployment of e-commerce solutions.
- Conducted system log analysis, debugging, and performance profiling; generated technical reports with actionable improvement recommendations.

FEATURED PROJECTS

AMR Warehouse Robot — Capstone Project · [Shifa National Medical Supply](#)

- Designed and deployed a production-grade autonomous mobile robot for intelligent warehouse management and medical supply distribution at a live healthcare facility.
- Implemented ROS navigation stack with RTAB-Map for real-time 3D SLAM mapping, autonomous path planning, and dynamic obstacle avoidance.
- Integrated YOLO-based computer vision pipeline for precise object recognition, inventory tracking, and automated picking assistance.
- Achieved 40% reduction in manual warehouse operations; delivered end-to-end system covering mechanical design, embedded firmware, and autonomous navigation software.

TECHTRAP — AI Rehabilitation & Educational System · [Huawei ICT Competition 2025, Top 5 National](#)

- Developed AI-powered therapeutic gaming platform for real-time neuro-motor assessment in children aged 4–12, requiring only a standard webcam with no specialist hardware.
- Built fully on-device inference pipeline using MediaPipe pose estimation and Huawei MindSpore neural network — zero cloud dependency, offline-first operation.
- Automated clinical report generation producing structured PDF assessments with performance charts and AI clinical interpretation in under 60 seconds per session.
- Designed adaptive game mechanics across 4 therapeutic domains; demonstrated live to competition judges, Ministry representatives, and medical professionals at national event.

FireX Robot — Autonomous Firefighter Robot · [Ministry of Interior & Civil Defense Partnership](#)

- Engineered fully autonomous firefighter robot for deployment in hazardous environments, demonstrated at the national fire festival in official government collaboration.
- Integrated ESP32-based embedded control with real-time thermal imaging, obstacle detection, and autonomous fire identification and suppression navigation.
- Developed MQTT-based real-time monitoring dashboard with remote override capability for safe operator control in high-risk environments.

Advanced Computer Vision Projects

- Built multiple real-time systems including object detection, gesture recognition, and visual analysis pipelines using OpenCV, PyTorch, and TensorFlow.
- Achieved 95%+ accuracy on custom datasets through deep learning model optimization and transfer learning techniques.

TECHNICAL SKILLS

AI & Machine Learning: Python, TensorFlow, PyTorch, Huawei MindSpore, Deep Learning, Neural Networks, Transfer Learning

AI Automation: LLM Integration, AI Agents, Prompt Engineering, Workflow Automation, API Orchestration, Process Automation

Computer Vision: OpenCV, YOLO (v8), MediaPipe, Pose Estimation, Object Detection, Image Processing, Real-time Inference

Robotics & Embedded: ROS (Noetic), SLAM (RTAB-Map, Nav Stack), Arduino, ESP32, Microcontrollers, Sensor Integration, MQTT

Programming: Python, C/C++, JavaScript, Java

Web Development: React, Node.js, Express, MongoDB, REST APIs, Shopify Liquid, Full-Stack Development

Tools & DevOps: Docker, Git/GitHub, Linux, Cloud Platforms

CERTIFICATIONS

- Gemini Certified University Student — Google
- Introduction to Modern AI — Cisco
- Internet of Things Fundamentals — Cisco
- Python Essentials 1 — Cisco
- Microsoft Professional Certificate — Microsoft

- Huawei AI Innovation Certificate — Huawei ICT Academy
- One Million Prompters — Dubai Centre for Artificial Intelligence

LANGUAGES

English: Fluent | Arabic: Native Speaker